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ENGINEERING

# SIMATS Online Programming Lab

## JAVA PROGRAMMING

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### CO3-AT3-Debugging and Optimization

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#### QUESTIONS

**Q1**ATM Withdrawal – Exception  
Handling Debugging  
10 marks**Q2**Hospital Registration –  
NullPointerException  
Debugging  
10 marks**Q3**E-Commerce Product Search –  
Array Exception Debugging  
10 marks**Q4**Bank Account – Race Condition  
Debugging  
10 marks**Q5**Railway Reservation –  
Synchronization Debugging  
10 marks

5/5 answered

**Q3**

#### E-Commerce Product Search – Array Exception Debugging

10  
marks

An online shopping application allows customers to select a product using a product number from 1 to 5.

The following program contains an array indexing error.

Task:

1. Identify the error.
2. Correct the product-number and array-index logic.
3. Handle invalid product numbers.
4. Handle non-numeric input.
5. Submit the COMPLETE corrected Java program.
6. The program must display "Search completed" after processing.

Faulty Program:

```
import java.util.Scanner;

class ProductSearch {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String[] products = {
            "Laptop",
            "Mobile",
```

```

"Tablet",
"Headphone",
"Keyboard"
};

System.out.print("Enter product number (1-5): ");
int productNumber = sc.nextInt();

System.out.println("Selected product: "
+ products[productNumber]);

System.out.println("Search completed");
}
}

```

Correct the program so that product numbers 1, 2, 3, 4 and 5 correctly access the corresponding products.

**🔧 Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

### Your Fixed Code

```

1 import java.util.Scanner;
2 class ProductSearch
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in)
7         String[] products = { "laptop", "mobi
8         System.out.println("enter product nu
9         try
10        {
11            int productnumber = sc.nextInt()
12            if(productnumber>=1 && productnu
13            {
14                int index = productnumber -1
15                System.out.println("seleced
16
17            }
18            else
19            {
20                System.out.println("invalid

```

```
21         }
22
23     }
24     catch (Exception e)
25     {
26         System.out.println("invalid input")
27     }
28     System.out.println("search completed")
29 }
30 }
```



### Test Input

3

### Output

```
enter product number(1-5):
selected prodducttablet
search completed
```

 Pending manual review

 Submitted